

Curriculum Vitae

Dilanjana Kaviraj Diyabalanage

PERSONAL INFORMATION

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EDUCATION

- **PhD Candidate in Physics** (Collaborative Specialization in Machine Learning in Health and Biomedical Sciences)
University of Western Ontario, 2022 – Present
Focus: Application of physics-based models and machine learning techniques, including Physics-Informed Neural Networks (PINNs), for the analysis and interpretation of neuroimaging data.
 - **MSc in Physics**
University of Western Ontario, 2021 – 2022
 - **B.Sc. (Hons.) in Physics (Special - 2nd Upper Division), GPA: 3.59**
University of Colombo, Sri Lanka, 2016 – 2020
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RESEARCH EXPERIENCE

- **PhD Research Assistant**
University of Western Ontario, 2022 – Present
Project: Enhancing Neuroimaging Analysis through Physics-based Models and Deep Learning
Responsibilities include developing and applying advanced computational models to interpret complex neuroimaging data, incorporating machine learning techniques to refine these models.
 - **Master's Research Assistant**
University of Western Ontario, 2021 – 2022
Project: Analysis of Source Reconstructed EEG data by a Model inspired by Quantum Mechanics
Modeled and reconstructed EEG data to extract dynamical features and relate them to spatial information, enhancing understanding of brain behavior.
 - **Undergraduate Research Assistant**
Department of Physics, University of Colombo, 2019-2022
Project Title: "Investigating Sonoluminescence Bubble Dynamics in Ionic Solutions"
Developed experimental setups to observe sonoluminescence—light emitted by collapsing bubbles in ultrasonic fields. Techniques included single and multi-bubble observations in various NaCl concentrations, dynamic radius measurements using laser scattering, and light intensity analysis via a photomultiplier and photon counting.
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TEACHING EXPERIENCE

- **Teaching Assistant**

Department of Physics and Astronomy, University of Western Ontario, 2021 – Present
Instructed laboratory courses covering motion, optics, electronics, and fluids. Conducted tutorials for first-year physics courses.

- **Teaching Assistant**

Department of Physics, University of Colombo, March 2020 – April 2021
Responsibilities included teaching optical experiments, circuit design, and PCB construction. Managed 120 freshmen and sophomores. Designed course modules for online laboratory sessions during the COVID-19 pandemic.

LEADERSHIP AND VOLUNTEER EXPERIENCE

- Member, Physics and Astronomy Graduate Council. 2022-Present
- Councillor, Society of Graduate Students. 2023 - Present
- Member, Sustainability Committee, Social and Orientation Committee, and International Graduate Students Issues Committee. 2023/24
- Volunteer and Instructor, Science Rendezvous, University of Western Ontario, 2022, 2023.
- Health Ambassador, University of Western Ontario, Fall 2021, Winter 2022.

AWARDS AND HONORS

- **Gulamhusein A. J. Noorbhai Gold Medal for the Best Physics Research**
University of Colombo, 2020

SKILLS

- **Programming and Software:** MATLAB, Mathematica, STATISTICA, ORIGIN, R Studio
- **Programming Languages:** Python, R, C, Java, PHP, JavaScript, LaTeX
- **Electronics:** Arduino, AVR programming, Autodesk Eagle, Multisim, Electronics Workbench
- **Project Experience:** Developed a website and database management system for a cab service (PHP, MySQL); Simulated electronic and mechanical systems (MATLAB Simulink); Interfaced sensor modules for data acquisition (AVR, Arduino).
- **Business and Entrepreneurship:** Graduation Student Innovation Scholar (Morrisette Institute for Entrepreneurship, Western Ivey) - 2024

LANGUAGES

- Sinhala (Native)
- English (Fluent, TOEFL - 106)

